CEE292 Project Proposal: Improved PINNs for Option Pricing

Sondre Rogde

Contribution: We implement the paper Y. Bai, T. Chaolu, S. Bilige, "The application of improved physics-informed neural network (IPINN) method in finance," *Nonlinear Dynamics*, 2022 in code. We implement both the standard PINN and their proposed improved PINN (IPINN).

Data and benchmark: No market data is needed. Training data consist of collocation, initial and boundary points sampled from the Black-Scholes and Ivancevic PDEs. We evaluate on a test grid against the closed-form Black-Scholes price and the analytic Ivancevic solution. We use relative L^2 error and maximum absolute error. We do this over 5 seeds and report mean $\pm$ standard deviation.